Estimating State Undocumented Populations from Published Survey Tables: A Reduced-Form Residual Method for Applied Policy Work

Eric A. Booth*

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Abstract

BACKGROUND. No public survey or administrative file enumerates people by undocumented status. The residual method, noncitizens minus a reconstructed legal noncitizen population, is the standard alternative, but published implementations need microdata and appear irregularly, leaving state analysts without current numbers.
OBJECTIVE. Show that state undocumented counts can be built from published American Community Survey (ACS) tables plus a few published anchors, and map where the shortcut is accurate.
METHODS. We take each state's noncitizen count from ACS table B05001, scale it by the national undocumented share of noncitizens (calibrated to the Center for Migration Studies (CMS) national total), and compare with CMS's state-specific residual estimates for 2024. We extend it to health insurance using table B27020.
RESULTS. The reduced form comes within 10 percent of the CMS state estimates in 22 of 51 states and within 20 percent in 36. Because both rest on the same ACS count, that agreement measures departure from the national share, not independent validation. Errors are systematic: the shortcut overestimates gateway states (New York +22%) and underestimates newer Southern destinations (Louisiana −30%). Small states drive a heavy tail: median absolute error is 12 percent, but 15 exceed 20 percent and two exceed 100.
CONCLUSIONS. A national ratio yields usable state levels but needs a local anchor where immigrant composition is unusual; the difference map shows where.
CONTRIBUTION. A transparent, published-tables-only residual estimator for every state, benchmarked against state-specific estimates, with a map of where a local anchor is needed and a microdata extension carrying quantified uncertainty.

Keywords: undocumented immigrants; unauthorized immigrants; residual method; American Community Survey; small-area estimation; state estimates; health insurance; reproducible research; immigration policy

*Correspondence: eric.a.booth@gmail.com

1 Introduction

No public data source enumerates undocumented immigrants directly. The ACS records citizenship, not legal status, and no complete public administrative file identifies everyone living in the country without full legal status. Two quantities can be measured: the total noncitizen population, which surveys capture reasonably well, and the legal noncitizen population, which can be reconstructed from administrative records. Residual estimates take the undocumented population as the difference between them. This logic has anchored unauthorized-population estimation since Warren and Passel (1987), and it remains the method behind the figures most cited in policy debate: Pew Research Center (Passel and Cohn, 2018; Passel and Krogstad, 2025), the Migration Policy Institute (MPI), the Center for Migration Studies (CMS), and the Department of Homeland Security's own Office of Homeland Security Statistics, which put the unauthorized population at 11.0 million as of January 2022 (Office of Homeland Security Statistics, 2024). These four implementations share the identity but differ in how they build the legal stock, so "the residual method" names a family rather than a single procedure (Duleep et al., 2025). Warren et al. (2023) contribute one recent refinement within that family, deriving the year-specific emigration rate from movement in the ACS rather than from a fixed historical constant; we follow their formulation here, though Pew and MPI use their own.

A larger disagreement sits outside the residual family altogether. Fazel-Zarandi et al. (2018) model inflows and outflows directly rather than differencing survey and administrative stocks, and arrive at roughly 16 to 29 million, well above the residual consensus near 11 to 15 million. The gap turns on what each approach treats as observable. The residual family trusts that the ACS counts most unauthorized residents, subject to a bounded undercount adjustment, while the inflow-outflow approach infers a population the survey never sees, from apprehension rates and border-crossing parameters estimated for earlier decades. Van Hook et al. (2021) test the residual side of that argument directly and find its total robust to reasonable variation in imputation, which is the main reason the demographic literature has not adopted the higher figures. We take the residual consensus as given here, and note that a reader who rejects it should reject this paper's levels along with the anchors they rest on.

These are careful constructions, and they are expensive to produce. Each pulls person-level microdata, imputes likely legal status case by case, rolls a legal population forward from Department of Homeland Security admissions and life tables, and adjusts survey counts for undercount. They also appear on their authors' schedules, not on a legislative calendar. A state agency that needs a current, defensible number for its own population, or a coverage estimate for a specific program, usually cannot rebuild that machinery on a deadline, and the last published state number may be several years old.

This paper offers a lighter route to state estimates by exploiting one practical feature of the residual identity: over short periods, the ratio of undocumented residents to all noncitizens changes more slowly than the annual demand for a current policy number. We call that ratio θ. It is what the published residual estimates implicitly pin down whenever they report a number for a place and year. Our method reads each state's noncitizen count from a published ACS table, sets θ from a published anchor, and multiplies. The estimate equals the anchor source's published number wherever an anchor exists, and follows the observed ACS count everywhere else.

The approach follows recent *Demographic Research* work that builds population estimates from published or auxiliary data and validates them against an independent benchmark. Abel (2013) infers migration flows from successive foreign-born stock tables; Acolin et al. (2022) calibrates small-area estimates from auxiliary data and checks them against the census; Stone (2023) and Stone et al. (2025) identify hard-to-measure subpopulations from ACS variables and defend the estimates against external registries; Hauer and Byars (2019) publish a transparent scripted pipeline as a reusable product; and Zoraghein and O'Neill (2020) fit and validate population models state by state. This paper follows that tradition: a parsimonious estimator built from public tables, checked against a state-specific residual estimate, with the code released.

Two lines of applied work have converged on a state-level undocumented count as their load-bearing input, and both currently rely on Pew or the Migration Policy Institute (MPI) directly, without a reproducibility trail an outside analyst can rerun. The first is enforcement-weighted labor-market and housing analysis, where county or metropolitan unauthorized shares weight treatment intensity in the wake of Immigration and Customs Enforcement (ICE) actions (Howard et al., 2026; Cox and East, 2026; Escobari et al., 2026; Cravino et al., 2026). The second is fiscal and gross-domestic-product (GDP) scoring split to states (Penn Wharton Budget Model, 2025; Congressional Budget Office, 2024; Institute on Taxation and Economic Policy, 2024), including 50-state projection exercises built directly on state counts (Kallick and Capote, 2024). The fit is partial rather than turnkey: the enforcement studies identify effects off county or metropolitan shares, and the fiscal models run microsimulation over person-level records, so neither takes a state total directly as its estimating input. What a published-tables estimator with an explicit error map offers both literatures is a transparent, annually refreshable state-level denominator for the places where they now cite Pew or MPI without a reproducibility trail, and a way to check how much their state-level inputs depend on the anchor source they happened to choose.

The contribution has three parts. We show that the residual method can be run for every state from published ACS summary tables and a single national anchor, without microdata. We then benchmark the shortcut against the Center for Migration Studies' state-specific published residual estimates for 2024 and map where it succeeds and where it fails, turning "is the shortcut safe here?" into a figure an analyst can read. And we work a microdata example that disaggregates the estimate to subgroups and sub-state regions and quantifies the added uncertainty with replicate weights, so users can judge when the extra effort is warranted. Throughout, the data are public, the steps are arithmetic, and the workflow is released.

The durable part is the method: a transparent shortcut that any analyst can apply to any subgroup, geography, or policy domain that has a published anchor and a survey count, packaged so it can be rerun as data arrive. The health-coverage estimates are one application of that method, and a deliberately narrower one, since they lean on a single insurance benchmark and a Texas-only calibration. A reader who wants a reusable tool should take the method; a reader who wants coverage numbers should take them with the uncertainty and scope caveats in Sections 5.5 and 10.

The rest of the paper is a working manual for that shortcut. We set out the residual identity and show that its central ratio θ moves slowly enough to interpolate between anchors (Section 2), pin the arithmetic to two public ACS tables (Sections 3–4), and run the shortcut for all 50 states plus the District of Columbia in 2024, benchmarking each against the CMS state residual and mapping where the shortcut is trustworthy (Section 5). Three applied vignettes and a Texas microdata build show what the method can and cannot do in practice (Sections 6–7). Reproducibility, implications, and limits follow (Sections 8–10), and a Texas-anchored time-trend sub-analysis closes (Section 11). By the end the reader can produce a defensible state undocumented count for their own jurisdiction from public inputs alone, and knows exactly when to hand the problem off to a heavier method.

2 The Residual Method and the Stability of θ

The identity. Write $U_{g,t}$ for the undocumented population of geography g in year t, $C_{g,t}$ for survey-measured noncitizens, and $L_{g,t}$ for the legal noncitizen population. The residual method begins from

$$U_{g,t} = C_{g,t} - L_{g,t}. \tag{1}$$

The survey term comes from the ACS, adjusted upward for undercount. The legal term is a stock that each source rolls forward year by year from arrivals (Department of Homeland Security lawful admissions), naturalizations, deaths (life tables), and emigration. Warren et al. (2023) estimate the emigration term from the year-over-year change the ACS shows in each cohort, which is the part of the accounting that had previously been fixed from 1980s data.

Why the ratio moves slowly. Define the undocumented share of noncitizens as

$$\theta_{g,t} \equiv \frac{U_{g,t}}{C_{g,t}} = 1 - \frac{L_{g,t}}{C_{g,t}}. \tag{2}$$

Each component that changes $L_{g,t}$ from one year to the next is a small fraction of the stock, and $C_{g,t}$ also changes gradually, so $\theta_{g,t}$ is a slow-moving demographic quantity. Over 2010 to 2023 Pew's national series implies a θ that drifts from 0.51 to 0.61, about three-quarters of a percentage point a year. We report the Pew series specifically rather than speaking of the sources collectively, because each source's θ path depends on its own legal-stock construction and we have verified the smoothness claim only for Pew, whose series has the most published anchor years. The holdout test in Section 5.3 bears directly on this assumption and covers CMS and MPI anchors as well. That smoothness is what the method exploits: if θ is known at a few points, its value nearby is well approximated. The same logic underlies stock-based flow estimation in demography (Abel, 2013) and survey-parsimonious reconstruction of migration trends (Schoumaker and Beauchemin, 2015).

3 Data

Who counts as undocumented. We follow the definition shared by the residual-method sources. The undocumented (unauthorized) population includes people who entered without inspection or overstayed a visa and people holding temporary or liminal protections: recipients of Deferred Action for Childhood Arrivals, holders of Temporary Protected Status, humanitarian parolees, and pending asylum applicants. This matters for interpretation. Pew reports that more than 40 percent of the 2023 population held some protection from deportation (about 6 of 14 million), and the Center for Migration Studies (CMS) reports that liminal-status immigrants were more than a third of its 2024 total (about 5.4 of 14.6 million).

Do the three sources define the same population? Yes, largely, and it matters because a reader might reasonably suspect the cross-source spread is definitional. Table 1 sets out how Pew, CMS, and MPI classify the groups whose treatment could plausibly differ. All three count the temporary and liminal statuses (DACA, TPS, humanitarian parole, pending asylum) as unauthorized and exclude lawful permanent residents, so the definitions line up. What does not fully line up is the *magnitude* each source assigns to some groups: Pew counts about 0.70 million parolees in the 2023 unauthorized population, while CMS's 2024 tool reports about 1.09 million. So part of the cross-source gap in Section 5.3 is compositional, not a pure difference in method.

ACS inputs. Two published ACS one-year detailed tables supply the inputs. Table B05001, Nativity and Citizenship Status, supplies total population and noncitizens for each state. Table B27020, Health Insurance Coverage Status and Type by Citizenship Status, supplies noncitizen

Table 1. How the three residual-method sources classify each group in their undocumented total.

Group	Pew	CMS	MPI	Why it matters
Entered without inspection	In	In	In	Core undocumented population
Visa overstays	In	In	In	Core undocumented population
DACA recipients	In	In	In	Temporary protection, not permanent status
TPS holders	In	In	In	Designations shift 2023–2024 totals
Humanitarian parolees	In	In	In	Grew sharply after recent parole programs
Pending asylum applicants	In	In	In	Largest liminal group ($\sim$2.6–3.6M)
Lawful permanent residents	Out	Out	Out	Legal noncitizen stock (L), subtracted

"In" = counted in the undocumented total; "Out" = excluded. Sources: Pew 2025 report and methodology; CMS State and National Data Tool ("About the data"); MPI methodology for assigning legal status (all accessed July 2026; URLs in Data Availability). DHS's Office of Homeland Security Statistics is a fourth residual source and reports only minor definitional differences from these three (Office of Homeland Security Statistics, 2024); we omit it from the columns because its most recent estimate has a January 2022 reference date and is not comparable to the 2023 and 2024 vintages compared here, and because it publishes no state detail. Pew treats *granted* asylees and refugees as lawful in their year of admission; only *pending* applicants are counted as unauthorized. The counts each source assigns to the liminal groups differ (for example, parolees about 0.70M in Pew versus 1.09M in CMS), so some of the cross-source spread is compositional.

counts of insured and uninsured people. We pulled both for all 50 states and the District of Columbia for 2024 through the Census Bureau API.[1] No person-level record is touched.

The anchor. The method needs, for each geography, the undocumented count from a residual source at one or more years. For the all-states 2024 estimate we use the CMS State and National Data Tool, which publishes a state-specific residual estimate for every state along with insurance detail. CMS is an immigration research institute; its national 2024 total is 14.61 million.[2] For the time-trend sub-analysis in Section 11 we additionally use Pew's national series and the Migration Policy Institute's Texas profiles as anchors; MPI's most recent national release puts the mid-2024 unauthorized population at 15.8 million (Migration Policy Institute, 2025), above both the Pew and CMS vintages used here, which is a further reminder that the choice of anchor source is itself a material part of the uncertainty.

[1] Retrieved with the Stata `getcensus` command and, in the released mirror, with a short Python API call. No microdata file is used.

[2] The CMS tool also underlies the 2023 cross-source comparison in Section 5.3, drawn from the Immigration Research Initiative (2025) 50-state compilation of CMS, Pew, and MPI estimates.

4 Methods

4.1 Reduced-Form Estimation

The estimator has three steps, each arithmetic. First, read $C_{g,t}$ directly from ACS table B05001. Second, set θ from a published anchor: at an anchor with published value $P_{g,a}$, $\theta_{g,a} = P_{g,a}/C_{g,a}$. Third, multiply:

$$\widehat{U}_{g,t} = \theta_{g,t} C_{g,t}. \tag{3}$$

By construction $\widehat{U}$ equals the published value wherever an anchor exists. Where none exists, the estimate carries the observed ACS count scaled by the nearest available θ. The implied legal population follows by rearrangement, $\widehat{L}_{g,t} = C_{g,t} - \widehat{U}_{g,t}$.

What we deliberately leave outside the estimator matters as much as what we build in. We do not rebuild admissions, life-table survival, emigration, or undercount adjustments; the published anchor already carries those adjustments, and our estimator inherits them the moment it borrows θ. The method inherits each source's demographic accounting at the anchor without rebuilding it, but it does not expose the components between or beyond anchors.

4.2 The All-States Application and the Two Methods Compared

For a single recent year, the cheapest anchor an analyst is likely to have is a national number. So we take the national undocumented share of noncitizens, θ_{nat}, and apply it to every state:

$$\widehat{U}_s^{\text{RF}} = \theta_{\text{nat}} \times C_s, \qquad \theta_{\text{nat}} = \frac{\sum_s P_s^{\text{CMS}}}{\sum_s C_s}, \tag{4}$$

where C_s is state s's ACS 2024 noncitizen count. Calibrating θ_{nat} to the CMS national total (0.599 in 2024) makes the reduced form sum to that total across states, so the state-by-state gap is purely the cost of assuming θ is uniform. This is **Method 1**. **Method 2** is CMS's state-specific published residual estimate, P_s^{CMS}, which lets θ vary by state.

The comparison answers a practical question. CMS publishes state numbers only sporadically, so in most years an analyst who wants a current all-states figure has only a national anchor and the annual ACS. Method 1 is what that analyst can produce. Method 2, where it exists, tells us how far Method 1 can be trusted, and the difference between them shows exactly where the uniform-θ shortcut breaks down.

4.3 Health-Insurance Transport

No survey identifies who is undocumented, so their uninsured rate cannot be read off the data directly; we build it from a rate that can. The ACS reports the uninsured rate for all noncitizens together, a group that mixes green-card holders, temporary legal residents, and the undocumented. Because undocumented residents are ineligible for most federally funded coverage and less likely to hold jobs that offer insurance, they are uninsured at a higher rate than legal noncitizens, so the combined rate understates them. We borrow one published estimate of the undocumented rate, measure how far it exceeds the observable noncitizen rate, and carry that gap to the years and places where only the noncitizen rate is available.

Formally, ACS table B27020 gives the noncitizen uninsured rate r_t^{NC}. For the all-states 2024 figures we take the uninsured and insured undocumented counts directly from the CMS tool, which reports them by state. For years or places without a published split, the workflow transports the noncitizen rate to the undocumented level on the *odds* scale, calibrated so that one anchor year matches a published benchmark rate (for Texas, MPI's 2023 rate of $1{,}306{,}000/1{,}966{,}000 = 0.664$). With $\text{odds}(r) = r/(1-r)$, the calibration constant and transported rate are

$$\kappa = \frac{\text{odds}(0.664)}{\text{odds}(r_{2023}^{NC})}, \qquad \widehat{r}_t^{U} = \frac{\kappa\,\text{odds}(r_t^{NC})}{1+\kappa\,\text{odds}(r_t^{NC})}, \tag{5}$$

and the counts follow as $\widehat{U}_t^{\text{unins}} = \widehat{U}_t\,\widehat{r}_t^{U}$ and $\widehat{U}_t^{\text{ins}} = \widehat{U}_t - \widehat{U}_t^{\text{unins}}$. We use the odds scale rather than a constant multiplier because a multiplier can push a rate above 1: constant-ratio calibration gives implausible 85 to 89 percent undocumented uninsured rates in the pre-ACA years, when noncitizen rates already exceeded 60 percent. The odds transform stays inside $(0,1)$, treats the insured and uninsured shares symmetrically, and reproduces the benchmark exactly at the anchor year. With $\kappa \approx 2.25$ the Texas transported rate is 79 percent in 2009 and 64 percent in 2024. The same equation transfers to another coverage or benefit domain given a domain-specific benchmark rate.[3]

5 Results

5.1 Twelve States Cover Three-Quarters of the 2024 Total

The reduced form places 14.6 million undocumented residents nationally in 2024, 4.3 percent of the population, matching the CMS total by construction. The state pattern is the familiar one

[3] Because the transport is anchored to MPI's Texas rate we report it for Texas; the national noncitizen uninsured rate (29 percent in 2024) would understate undocumented uninsurance. Section 5.6 tests this single Texas calibration against CMS's published state uninsured splits for all 50 states.

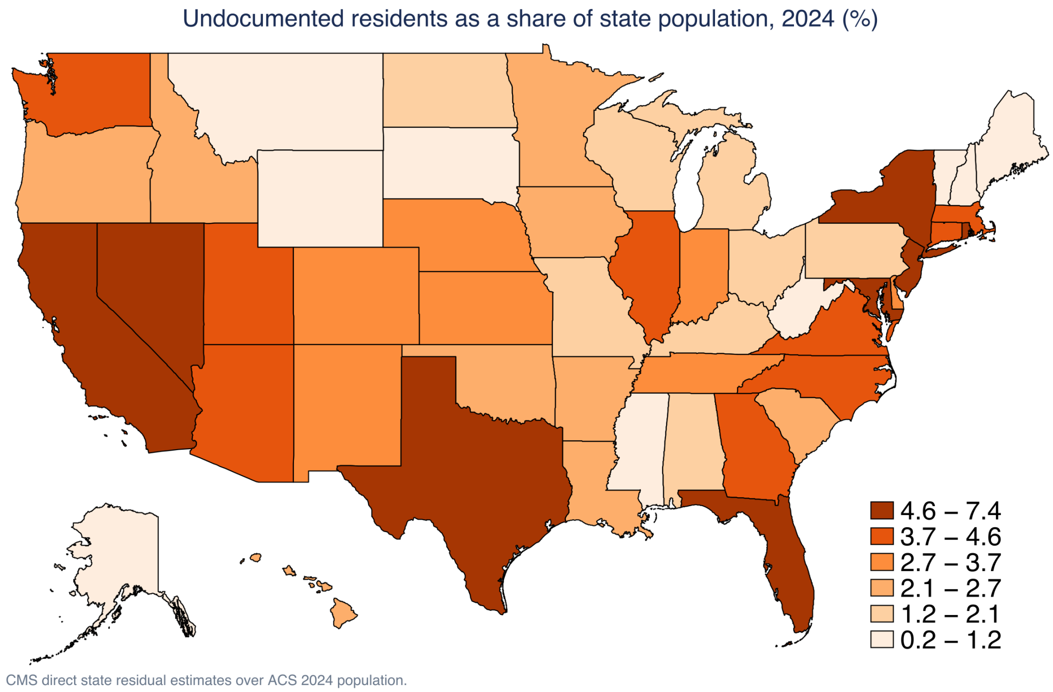

Figure 1. Undocumented residents as a share of state population, 2024. CMS state residual estimates over ACS 2024 population.

(Figure 1): the highest population shares fall in Texas and New Jersey (both about 7.4 percent), followed by California, Florida, and Nevada, near or above 6 percent, and the lowest in the Upper Midwest and northern New England. Table 2 lists the twelve largest state counts, which together account for about three-quarters of the national total.

5.2 Where the Reduced Form Agrees With the CMS Residual

Before reading the comparison, it helps to be exact about what it can and cannot establish. Both quantities begin from the same ACS noncitizen count: the reduced form is $\theta_{\text{nat}} \times C_s$, and the CMS state figure is that source's own θ_s applied to the same C_s. The comparison therefore measures how far each state's undocumented share of noncitizens departs from the national one. It is a diagnostic of θ heterogeneity, not a test against an independently observed count, and no such count exists. What it does establish is practical and worth having: whether an analyst who has only the national ratio would arrive at a number close to the one a full residual construction would have produced.

The reduced form reproduces the country total by construction and is close for many states, but for a specific state it can be off by a good deal, and that gap, not the national aggregate, is what a state analyst needs to know. Figure 2 plots Method 1 against Method 2 on log scales; points cluster along the 45-degree line, and the largest states sit almost on it. The reduced form falls within 10 percent of the CMS state residual in 22 of 51 states and within 20 percent in 36 of 51. The national

Table 2. Twelve largest state undocumented populations, 2024. Reduced form is Method 1 (national $\theta\times$ ACS noncitizens); CMS residual is Method 2.

State	ACS noncit. (M)	Reduced form (M)	CMS direct (M)	Diff. vs CMS (%)	Undoc. % of pop.	Sources ’23 range (M)
California	5.00	3.00	2.63	+14	6.7%	2.25–2.91
Texas	3.35	2.01	2.32	-13	7.4%	1.97–2.05
Florida	2.43	1.46	1.43	+2	6.1%	1.03–1.60
New York	1.88	1.12	0.92	+22	4.6%	0.78–0.84
New Jersey	1.05	0.63	0.70	-11	7.4%	0.48–0.60
Illinois	0.95	0.57	0.58	-3	4.6%	0.47–0.59
Georgia	0.70	0.42	0.50	-17	4.5%	0.41–0.48
North Carolina	0.63	0.38	0.46	-18	4.2%	0.37–0.45
Washington	0.64	0.38	0.37	+5	4.6%	0.33–0.38
Virginia	0.53	0.32	0.34	-7	3.9%	0.30–0.36
Maryland	0.48	0.29	0.33	-11	5.2%	0.31–0.37
Massachusetts	0.64	0.38	0.32	+19	4.5%	0.27–0.40
United States	24.40	14.61	14.61	0	4.3%	12.24–14.00

Diff. is the reduced form relative to the CMS residual. Undoc. % of pop. and the CMS/Pew/MPI columns use published 2024 (CMS) and 2023 (sources) figures. “Sources ’23 range” is the lowest to highest of the three published estimates (a robustness range, not a confidence interval; Section 5.5); ACS sampling error on the reduced form is separate and small (about 1–2% for these states). CMS rounds state counts to the nearest thousand.

totals agree by construction, and the two state distributions differ by only about 6 percent overall (a standard dissimilarity index), so in aggregate the shortcut is close. The per-state errors, however, are several times that and follow a clear regional pattern.

The divergences are not random, which is what makes them useful. Figure 3 maps the percentage difference. The reduced form overestimates in gateway and high-legal-immigration states, where many noncitizens are green-card holders, students, or temporary workers and the undocumented share of noncitizens sits below the national average: New York (+22%), California (+14%), and Massachusetts (+19%). It underestimates across the newer Southern and Mountain destinations, where the undocumented share runs high: Louisiana (−30%), Tennessee (−23%), Utah (−23%), North Carolina (−18%), Georgia (−17%), and Texas (−13%). The pattern is a single fact seen two ways: the error in each state is exactly how far its own θ sits from the national θ. Because immigrant composition is regionally structured, the errors are too.

The headline agreement counts hide a heavy tail, so here is the full error distribution. The median absolute error across the 51 units is 12 percent, the 75th percentile is 23 percent, and the 90th is 40 percent. Fifteen states exceed 20 percent, four exceed 50 percent, and two exceed 100 percent: Vermont, where the reduced form is more than five times the CMS figure (+432%),

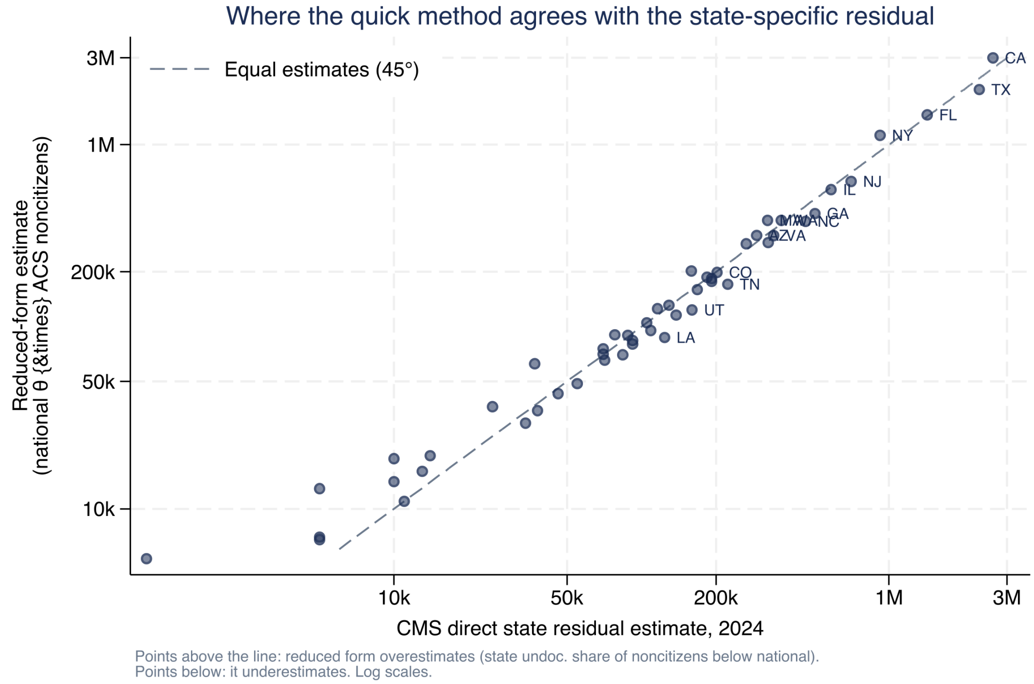

Figure 2. Reduced form (Method 1) versus CMS residual (Method 2), 2024, log scales. Points above the 45-degree line are states where the reduced form overestimates.

and Alaska (+158%), followed by New Hampshire (+88%) and Hawaii (+69%). These are the smallest noncitizen populations in the country, where CMS rounds to the nearest thousand and the ACS samples thinly, so a few hundred people swing the ratio. The shortcut should not be used for them at all without a state-specific anchor. Naming them matters more than averaging them away: an analyst who reads only the aggregate agreement statistics would not learn that the method fails outright in a handful of small states.

The stakes elsewhere are concrete. An analyst who applied the national ratio to Louisiana would understate its undocumented population, and with it the uninsured count and the scope of any coverage program, by about a third; the same shortcut would overstate New York by a fifth. Read the map as a decision tool, not a diagnostic curiosity: it tells an analyst which states a quick national estimate can be trusted for and which need a state-specific anchor first.

5.3 θ Interpolates Cleanly, and Cross-Source Estimates Broadly Agree

Two checks test the method against numbers it was not handed. The first is a leave-one-anchor-out holdout on the interpolation step: for each series with at least three anchors we drop each interior anchor, re-interpolate θ from the remaining anchors, predict the held-out year, and compare to the source's published value (Table 3). Across eleven held-out anchors the mean absolute error

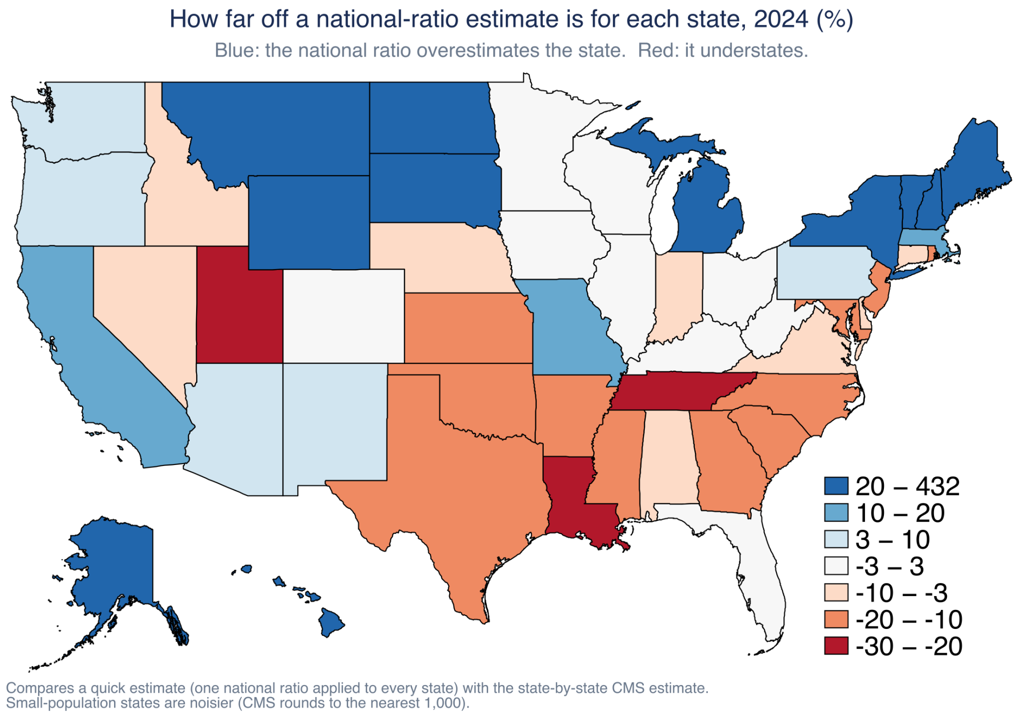

Figure 3. How far off a national-ratio estimate is for each state, 2024, percent. Blue states are overestimated by the national-ratio shortcut; red states are underestimated. Small-population states are noisier because CMS rounds to the nearest thousand and ACS sampling error is larger there.

is 2.4 percent, and the worst miss is 5.0 percent (Pew's 2021, the pandemic dip year, which any smooth interpolation will struggle with). That is direct evidence that θ moves smoothly enough to interpolate.

The second check uses 2023, the one recent year for which Pew, MPI, and CMS all publish. Table 4 shows the national spread is about 14 percent (CMS 12.2 million, MPI 13.7 million, Pew 14.0 million). The reduced form reproduces whichever anchor it is given, so its role here is to keep that cross-source range in view: any single anchor an analyst adopts, whether the CMS low end or the Pew high end, sits within a band about 14 percent wide nationally and 4 percent wide for Texas, and the choice of source is itself a meaningful part of the uncertainty.

That band widens once the federal estimate is included. DHS's Office of Homeland Security Statistics puts the unauthorized population at 11.0 million for January 2022 (Office of Homeland Security Statistics, 2024), below all three of the 2023 figures. Part of that gap is the reference date, since the 2022–2023 period saw the sharpest recent growth, so the comparison is not like-for-like and we keep OHSS out of Table 4. But an analyst choosing an anchor in practice chooses among sources with different reference dates as well as different methods, and the federal series sits at the low end of that choice.

Table 3. Leave-one-anchor-out validation of the θ-interpolation step. Each interior anchor is dropped, predicted from the others, and compared to the source's published value.

Series	Held-out year	Published (millions)	Predicted (millions)	Error
U.S., Pew	2015	11.00	10.78	-2.0%
U.S., Pew	2017	10.50	10.79	+2.8%
U.S., Pew	2019	10.20	10.44	+2.3%
U.S., Pew	2021	10.50	11.02	+5.0%
U.S., Pew	2022	11.80	12.00	+1.7%
U.S., Warren/CMS	2019	10.35	10.70	+3.4%
U.S., Warren/CMS	2021	10.30	10.48	+1.8%
U.S., Warren/CMS	2022	10.90	11.06	+1.5%
U.S., Warren/CMS	2023	12.24	12.59	+2.9%
Texas, Warren/CMS	2018	1.73	1.75	+1.1%
Texas, Warren/CMS	2023	2.05	2.10	+2.5%

Mean absolute error 2.4%; median 2.3%.

Table 4. 2023 national and Texas estimates across residual-method sources, millions.

Source (2023)	United States	Texas
CMS (Warren)	12.24	2.05
Pew	14.00	2.05
MPI	13.74	1.97

Compiled by Immigration Research Initiative (2025). The spread is about 14 percent nationally and 4 percent for Texas.

5.4 A Components Reconstruction Reproduces the Legal Stock but Not the Residual

The reduced form borrows θ rather than rebuilding the legal population from its demographic parts; a components reconstruction is a useful check on what the parts give on their own. We implement the year-by-year Warren accounting for the United States directly: starting from the Department of Homeland Security's published 2015 lawful-permanent-resident (LPR) stock, we roll it forward with real DHS flows, $L_t = L_{t-1} + \text{admissions}_t - \text{naturalizations}_t - \text{deaths}_t - \text{emigration}_t$, using annual LPR admissions and naturalizations from the DHS Yearbook, a life-table mortality rate near 0.5 percent per year (consistent with the LPR age structure), and an emigration rate near 1 percent per year. Warren et al. (2023) derive the year-specific emigration rate from ACS movement, which is their central innovation; a flat rate is a deliberate simplification here. The reconstruction reproduces DHS's own published LPR stock to within about 2 percent across 2015–2024 (Figure 4, left), and the fit is stable across mortality rates of 0.4 to 0.7 percent and emigration rates of 0.5 to

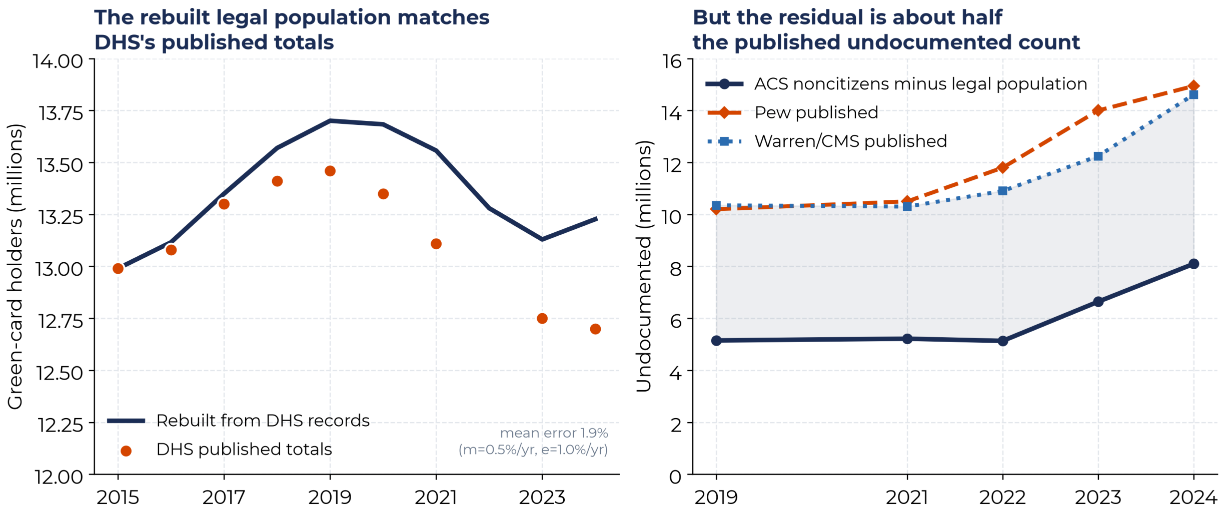

Figure 4. A components-based reconstruction of the U.S. legal noncitizen population. Left: rolling the LPR stock forward from DHS admissions and naturalizations, with life-table mortality and a 1 percent emigration rate, reproduces DHS's published LPR stock to within about 2 percent. Right: composing the undocumented as ACS noncitizens minus the DHS legal population yields roughly half the published Pew and CMS counts, because the administrative LPR stock exceeds the ACS-resident legal population, which is the coverage adjustment the anchor sources embed.

1.5 percent. That last point cuts both ways: a threefold range in the emigration rate leaves the fit intact, which means this exercise confirms the bookkeeping is internally consistent but does not pin down emigration itself.

Composing the undocumented residual from these parts is another matter. Taking the undocumented as ACS noncitizens minus the DHS legal population (LPRs plus the resident nonimmigrant population) gives about 5 to 8 million for 2019–2024, close to half the published Pew and CMS figures of 10 to 15 million (Figure 4, right). The reason is not an error in the flows but a level difference: DHS's *administrative* LPR stock (near 13 million) exceeds the *ACS-resident* legal noncitizen population, because administrative records shed people who have emigrated or died more slowly than a survey counts them. Closing that gap is the emigration-and-coverage reconciliation that the anchor sources embed in their published U_t, and that the reduced form inherits the moment it borrows θ. The two panels establish different things. The left is an internal consistency check, since DHS-based flows and a DHS-based stock come from the same administrative system. The right is a measurement: it sizes the coverage adjustment at roughly half the published total. The exercise therefore does not validate the reconciliation; it shows how much work that step is doing, and how much of the published estimate rests on a step we take on faith from the anchor. For applied work that argues for borrowing θ rather than rebuilding it, and Section 10 flags it as the method's largest untested assumption.

5.5 How Uncertain Are the Estimates?

The estimates carry two very different kinds of uncertainty, and we keep them apart rather than blending them into one interval. The first is ordinary ACS sampling error, a genuine sampling margin. For the reduced-form state estimate it is small: about 1 to 2 percent for large states (a 90 percent margin of ± 1.6 percent for Texas, ± 1.1 percent for California) and a few percent for the smallest. We report it as the margin of error and carry it in the released file.

The second, and much larger, is disagreement among the published sources: for the same state and year, how far apart are CMS, Pew, and MPI? We report this as a *range*, not a confidence interval, and the distinction is deliberate. Three estimates from different organizations, methods, and reference years are not a sample from a distribution; with only three, non-independent (they share ACS data and the residual logic) and partly separated by vintage, their spread is a robustness range, not a variance one can defensibly turn into a 90 percent bound. Read that way it is large and uneven (Figure 5 and the last column of Table 2): the three sources span a median of about 26 percent of their mean across states, from near-agreement in Texas (1.97 to 2.05 million) to a 2.5-fold gap in Michigan (0.08 to 0.20 million). Our reduced-form estimate falls inside the published range in about half the states; where it falls outside it does so in the direction the difference map already flags, below the band in high-share states like Texas and above it in low-share ones.

The size of that spread tracks one thing: the undocumented share of a state's noncitizens. The residual method estimates the undocumented as a difference, noncitizens minus the legal population. Where they are a large share (Texas, $\theta \approx 0.69$) the difference is a robust quantity the three methods pin down similarly; where they are a smaller share of a pool heavy with international students, work-visa holders, and refugees (Michigan, $\theta \approx 0.47$) the same methodological choices swing a much thinner residual. Part of the widest gaps is also vintage, not method: CMS and Pew use the 2022 ACS while MPI pools 2019–2023, so a state's range mixes reference-period differences with method, and same-vintage sources would agree more closely. The disagreement is real and belongs in view, but it is a spread across methods, not a sampling interval.

A fully probabilistic interval on the undocumented count is possible but requires simulating within a single method's imputation, as Van Hook et al. (2021) do; that is beyond a reduced form built on published tables. And none of the three quantities here—the sampling margin, the source range, or the uniform-θ bias of Section 5.2—includes the ACS undercount adjustment already embedded in θ, so each is a component of uncertainty, not the whole of it.

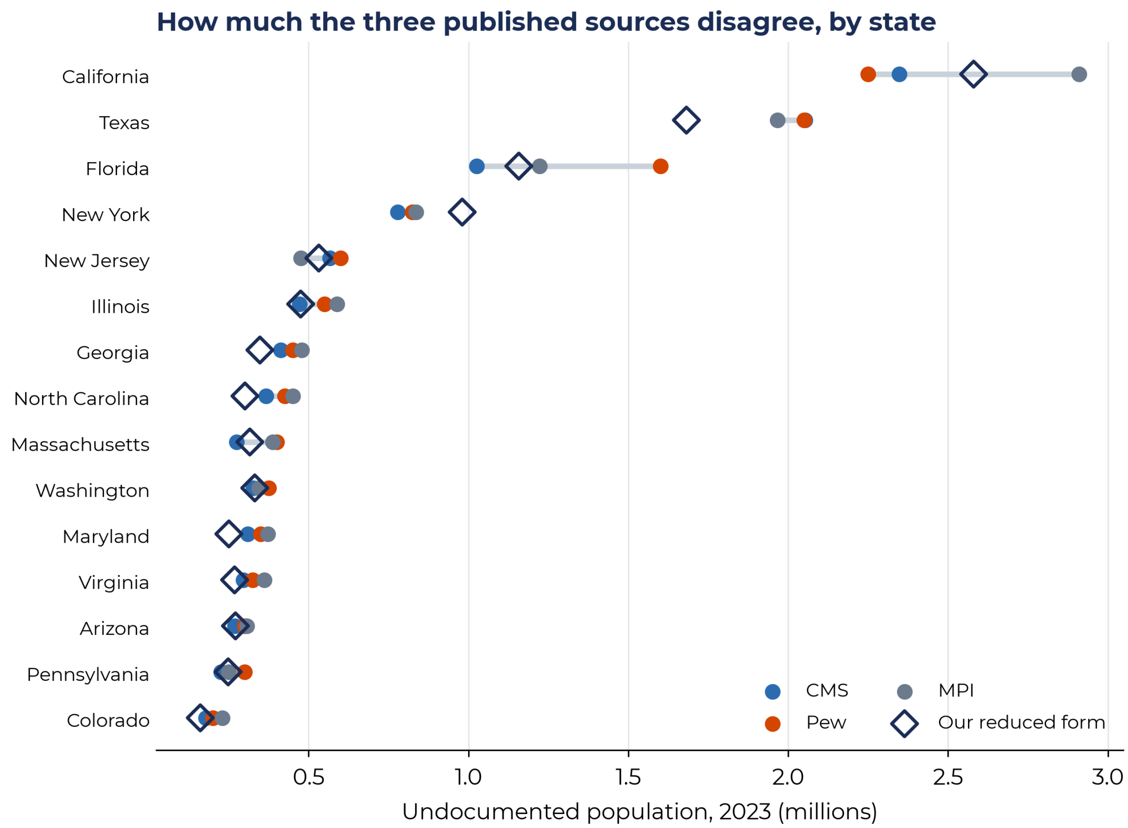

Figure 5. The three published residual-method sources (CMS, Pew, MPI) for 2023, by state, with our reduced-form estimate for comparison. The gray bar is each state's min-to-max source range. Disagreement is small in Texas and wide in states like Florida; our shortcut sits within the published range in about half of states.

5.6 Uninsured Undocumented Residents Concentrate in the Non-Expansion South

Coverage is where these counts most often meet policy, because undocumented residents are ineligible for most federally funded insurance and figure heavily in uncompensated-care and state-coverage debates (Hamilton et al., 2022). Nationally, CMS puts 6.6 million undocumented residents, 45 percent, without any coverage in 2024, a figure consistent with Artiga and Diaz (2019), who put 45 percent of nonelderly undocumented immigrants uninsured in 2017. The two are not independent confirmations of each other: they are seven years apart, cover different age ranges, and both rest on the ACS. The agreement is reassuring about order of magnitude and no more. The uninsured rate varies widely across states (Figure 6), from under 30 percent in states with broader coverage policy to near 60 percent in much of the South. The largest uninsured undocumented populations are in Texas (1.39 million, 60 percent uninsured), California (0.77 million, 29 percent), Florida (0.57 million, 40 percent), New Jersey (0.37 million), and New York (0.31 million).

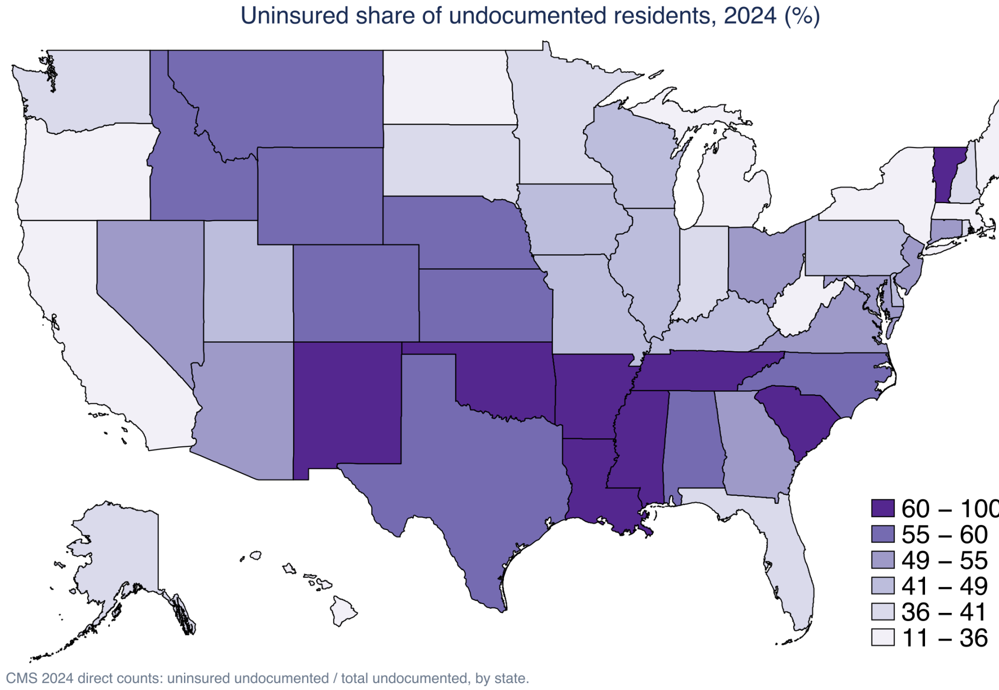

Figure 6. Uninsured share of undocumented residents by state, 2024. CMS 2024 counts: uninsured undocumented over total undocumented.

Texas and California have broadly comparable undocumented populations (California is somewhat larger) but very different uninsured shares, 60 against 29 percent, a contrast that reflects coverage policy rather than demography.

The insurance transport is calibrated only once, to MPI's 2023 Texas benchmark, so a fair question is whether that single calibration travels. Figure 7 tests it: for every state we transport the ACS noncitizen uninsured rate to the undocumented level with the one Texas-calibrated constant and plot it against CMS's directly published state uninsured rate. The points track the 45-degree line (correlation 0.71; a population-weighted mean error of about 3.7 percentage points), but the fit is loose: a correlation of 0.71 leaves about half the cross-state variance unexplained, which is what a constant calibrated in a single state should be expected to deliver. The residual variation is not random either, since Medicaid expansion status and state-funded coverage programs move the undocumented uninsured rate in ways no Texas-derived constant can anticipate. The transported rate is therefore serviceable where no published split exists, and should be replaced by the published state rate wherever one is available, as it is for all 50 states in the 2024 CMS release used in Section 5.6. This is a check on the rate transport, not a second data source. For Texas two benchmarks disagree by 6 percentage points — MPI's 2023 calibration gives 66 percent, CMS's published 2024 rate 60 percent — and we report the range rather than choosing between them.

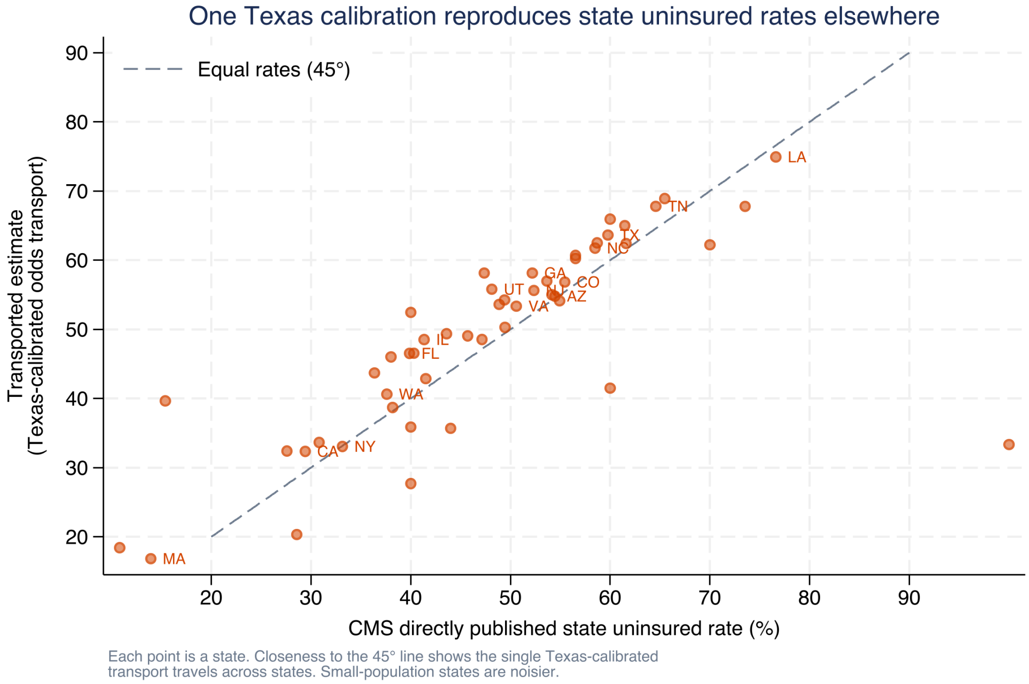

Figure 7. The single Texas-calibrated odds transport, applied to every state's ACS noncitizen uninsured rate, against CMS's directly published state uninsured rate, 2024. Closeness to the 45-degree line shows the calibration travels across states.

6 Worked Applications

This section works through the applied questions the method is built to answer. Each takes a single state as its example, shows what the shortcut returns, and marks where the difference map tells the analyst to adjust.

Application 1. When only a national anchor is available, the difference map still guides the report. Suppose CMS has not yet released state numbers for the latest ACS year, and an analyst in North Carolina needs one. Read North Carolina's ACS noncitizen count, apply the national θ of 0.599, and report the product. The difference map (Figure 3) warns that North Carolina runs about 18 percent above the national ratio, so the honest report reads "roughly 0.38 million on the national ratio, and likely higher, since North Carolina's undocumented share of noncitizens exceeds the national average." The shortcut gives a defensible starting number and, through the map, an informed direction of adjustment.

Application 2. Two residual constructions bracket a defensible range. A single point invites false precision. The reduced form and the CMS residual estimate together bracket a state: for Georgia, 0.42 million (national ratio) to 0.50 million (CMS residual), reported as "roughly 0.42 to 0.50 million from two residual constructions." A band built from two real methods is more defensible than a single point estimate, and it is available from the same inputs.

Application 3. Coverage-program scope, without touching microdata. A state agency projecting uncompensated-care costs or emergency Medicaid caseload needs the uninsured undocumented population, not just the total. Texas, at 1.39 million uninsured undocumented residents, and California, at 0.77 million, face problems of different size and shape despite broadly comparable totals. Table B27020 plus the CMS uninsured split delivers that number for any state without touching microdata. The same per-capita-multiplier logic extends to fiscal quantities: a state's undocumented count times a published per-capita state and local tax-contribution rate (Institute on Taxation and Economic Policy, 2024) recovers a state fiscal-contribution figure of the kind that Penn Wharton Budget Model (2025) and Congressional Budget Office (2024) attach to their national scenarios, with the difference map showing where the national-ratio version needs a state anchor first.

7 What ACS Microdata Would Add

The published tables give counts, not person-level detail, and most of the questions that follow the headline number are about people. How many of a state's uninsured undocumented residents are children? How many have lived in the country long enough to meet a residency rule? Where within the state do they live? No published table answers these; the person-level ACS Public Use Microdata Sample (PUMS) does. Moving to PUMS adds resolution, but it also adds cost and uncertainty, and the two are easy to confuse.

What it adds. With microdata the estimate can be disaggregated by age, country of origin, year of entry, income, education, language, and by geography below the state, down to the public use microdata area. That is the difference between a headline of 1.4 million uninsured undocumented Texans and the share of them who are children, or who have been in the country long enough to meet a residency rule. Those are the breakdowns that decide whom a given program actually reaches, and demography has standard tools for recovering such age-and-origin structure when only totals are observed (Raymer et al., 2025).

Person-level legal status. The deeper use of microdata is to stop treating θ as a single number and instead give each ACS noncitizen a probability of being undocumented. The usual way to do this borrows status-related information from a survey with richer immigration measures, historically the Survey of Income and Program Participation, and carries that information onto the ACS records, either through a set of logical edits and selection ratios (Warren, 2014; Borjas, 2017) or, more defensibly, through multiple imputation (Capps et al., 2018). This is how the Migration Policy Institute and the Center for Migration Studies build their state and local profiles, and it is the person-by-person version of the single ratio we apply in aggregate. There is a close model for it in this journal: Hamilton et al. (2022) carry legal status onto foreign-born survey records precisely to study who is insured, the same coverage question we answer here one state at a time.

What it costs. More detail is not the same as more certainty, and it is easy to take one for the other. Legal status is not observed even in the survey it is borrowed from, so it has to be modeled before it can be carried over (Bachmeier et al., 2014). A single pass of logical edits is quick but can be biased; multiple imputation does much better, but it carries an honest imputation variance that a single assignment simply hides (Van Hook et al., 2015). So a person-level estimate is richer and, cell by cell, often shakier. Even the national total has a wide range in the best current work (Van Hook et al., 2021), and a subgroup within a small state is thinner still, with sampling error and imputation error stacking on top of each other. In that regime the honest output is a range, not one imputed count (Goes and Engelhardt, 2026).

7.1 Worked example: Texas from the microdata

To make this concrete, we ran it. We took the 2024 one-year ACS PUMS for Texas, 292,272 person records, flagged noncitizens from the citizenship variable, and carried the same two numbers used throughout this paper down to individual cells: the undocumented share of noncitizens θ and the odds-scale insurance transport. These are *transported* estimates: the aggregate θ and the odds rate transform applied to finer cells. They are not person-level legal-status imputations, which would assign each ACS record a probability of being undocumented (the heavier build described above) and carry the extra imputation variance that goes with it. Every cell therefore inherits the statewide θ unchanged. Sampling uncertainty for each cell comes from the 80 PUMS replicate weights, using the successive-difference formula the Census Bureau supplies for this file. The microdata reproduce the aggregate before disaggregation: the weighted noncitizen count is 3.36 million against the table's 3.35 million, the implied undocumented total is 2.10 million, and the noncitizen uninsured rate is 0.44, as in table B27020. That agreement is what licenses the disaggregation.

One point of bookkeeping, because the numbers will look inconsistent otherwise. This example anchors on the Pew and MPI vintage, which puts Texas at 2.10 million for 2024 and implies $\theta = 0.63$, for continuity with Section 11. Section 5 anchors on CMS instead: 2.32 million, $\theta = 0.69$. The two differ by about 10 percent, well inside the cross-source spread of Section 5.5, and the choice scales every cell in Table 5 proportionally. A reader preferring the CMS vintage can multiply the counts by 1.10; rates and cross-cell comparisons are unaffected.

Table 5 shows what the statewide number hides. The transported coverage rates have a sharp age gradient: working-age undocumented adults are uninsured at 64 to 68 percent, but those 65

Table 5. Texas undocumented estimates disaggregated from the 2024 ACS PUMS. Undocumented is θ times the PUMS noncitizen count ($\theta = 0.63$); the uninsured rate is transported to the undocumented level on the odds scale. These are transported estimates, not person-level legal-status imputations.

Subgroup (Texas noncitizens, 2024)	Noncit. (000s)	Undoc. est. (000s)	±90% (000s)	Uninsured rate	Uninsured (000s)
All noncitizens (validation)	3,357	2,102	35	64%	1,339
Children 0-17	367	230	12	65%	149
Adults 18-34	926	580	16	68%	397
Adults 35-49	1,143	716	14	65%	468
Adults 50-64	662	415	10	64%	264
Adults 65+	259	162	7	28%	46
Adults 18-64 with a child under 6	685	429	19	68%	293
Adults 18-64 with a child 6-17	825	517	16	68%	352
Houston (Harris Co.): adults with a child	133	83	8	64%	53
Houston (Harris Co.): children 0-17	35	22	4	59%	13
Hidalgo Co. (Rio Grande Valley): adults with a child	74	46	5	76%	35

"Adults with a child" are noncitizen adults aged 18–64 living in a household with a child under 18, split by the age of the youngest child, so the two rows do not overlap. The two sub-state rows are single counties selected by public use microdata area: Harris County for Houston, and Hidalgo County, the most populous of the four Rio Grande Valley counties, for the border region. Neither covers its full metropolitan area or region. Counts in thousands. The ±90% column is the 90 percent ACS sampling margin on the undocumented estimate, from the 80 replicate weights (successive-difference replication); cross-source and undercount uncertainty are separate, as in Section 5.5. **The 65-and-older row is not comparable to the others and should not be quoted as a count.** Every cell here applies the same statewide $\theta = 0.63$, but noncitizens aged 65 and older have had far longer to naturalize or adjust status, so their true undocumented share is almost certainly well below the statewide figure and this row overstates the level. It is reported to show the age gradient in the transported coverage rate, not to size the older undocumented population.

and older at 28 percent. Because legal status is not observed in PUMS, the older-adult estimate should be read as a transported subgroup rate rather than direct evidence of coverage among undocumented older adults. The groups that drive mixed-status-family policy are large and mostly uninsured: an estimated 429,000 undocumented adults have a child under six at home, and a further 517,000 have school-age children and none under six, both near 68 percent uninsured. The two groups split on the age of the youngest child, so they do not overlap. And the geography lands where a state analyst would look. Houston (Harris County) has about 83,000 undocumented adults living with children; Hidalgo County, the most populous of the four Rio Grande Valley counties, has about 46,000, but their uninsured rate of 76 percent runs well above the state's 64 percent. That kind of local contrast can inform clinic siting and coverage planning, and the statewide total hides it.

None of this detail is free, and Figure 8 shows the cost as a plain sampling margin of error. As the cells shrink, the margin widens: about ±2 percent for the statewide estimate, around ±10 percent for a county subgroup, and ±20 percent for undocumented children in a single county, where the estimate of roughly 22,000 could be off by a fifth in either direction. That last number

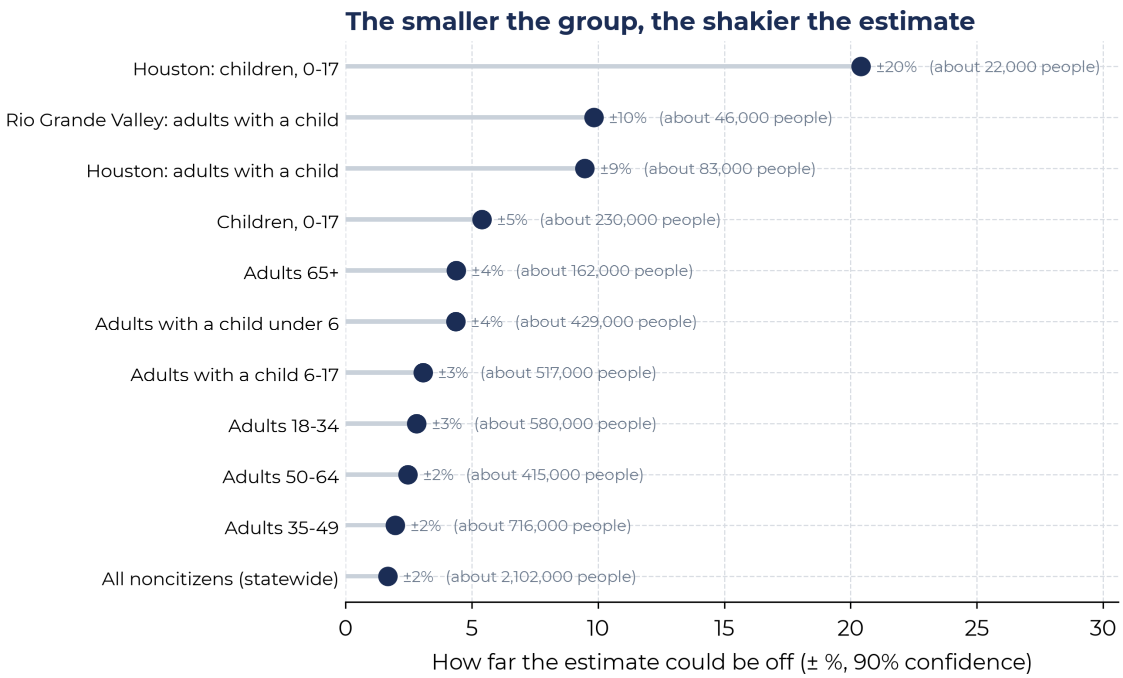

Figure 8. The smaller the group, the shakier the estimate. Each bar is a row of Table 5; its length is the 90 percent ACS sampling margin (from the 80 replicate weights) on the undocumented estimate. The statewide estimate is precise (± 2 percent); undocumented children in a single county reach ± 20 percent.

is usable for planning but should not be quoted as a point. In that specific sense the microdata strengthen the estimate: they answer questions the tables cannot, and they make the sampling uncertainty something we measure rather than assume.

That sampling error is only a floor, which is the honest catch. Every cell also is subject to the cross-source disagreement in θ, which adds another two to four percent, and a full person-level build would replace our single θ with an imputed legal status whose imputation variance, for the smallest cells, would likely swamp the sampling error (Van Hook et al., 2015, 2021). The lesson is not that microdata are unreliable but that detail and uncertainty grow together. PUMS is the right tool when a decision turns on a subgroup or a sub-state number, and a less useful tool when a statewide total would have served, because there it can add noise to a figure the published tables already pin down.

When it is worth it. The question is not which method is better but which question is being asked. For a current-year state total or an aggregate coverage count, the published tables and an anchor already answer it, and microdata would add work and noise without moving the number. For targeting, who qualifies, where they live within a state, how long they have been present, the tables cannot answer at all, and the extra uncertainty of the microdata build is a price worth paying. The two are complements rather than rivals: the reduced form gives fast, transparent levels, and the microdata gives structure when a particular decision turns on it.

8 Reproducibility

Because the whole appeal of the method is that anyone can run it, the workflow is written to be rerun, in the spirit of the replicable pipelines this journal treats as the default (Hauer and Byars, 2019). Everything starts from public inputs: the ACS tables come from the Census Bureau API, and the CMS state file is a public download. A single script turns those two inputs into the all-states estimates and every figure in Section 5, and a companion pipeline reproduces the time-trend results. Two habits make the output checkable rather than merely available. The estimate equals each published anchor at its anchor year by construction, and a validation file records a deviation of zero at every anchor, so a reader can confirm the method did not drift from the numbers it claims to reproduce. Every anchor value, its source, and the date we verified it are also written to a plain-text log, so the provenance of each input is visible instead of buried in code. Inputs and outputs are CSV throughout, and adding a new ACS year or a newly published anchor is a one-line edit and a rerun.

Running the method yourself. The calculations are arithmetic and need no special software, but we package them as a Stata command, `residualundoc`, for readers who would rather not rewrite them.[4] Two of its subcommands do the work described above. Given a variable holding each state's ACS noncitizen count and a variable holding the published CMS state estimates, `residualundoc apply noncit, direct(cms) tolerance(10)` reproduces the all-states comparison in Section 4.2: it calibrates θ_{nat} as $\sum_s P_s / \sum_s C_s$, applies it to every state, and returns the share of states within 10 percent along with the dissimilarity index. For a time series, `residualundoc interp year noncit, anchors(2010=11400000 2023=14000000)` sets θ at each anchor year and interpolates between, as in Section 11. The notation maps directly: C is the noncitizen variable, P enters through `direct()` or `anchors()`, and θ is returned in `r(theta)` or written to a variable.

Two cautions apply to anyone running it on new data. First, `anchors()` expects published undocumented *counts*, not shares; passing 0.61 where 14000000 belongs is the most common way to get a silent nonsense answer, and the command now refuses that input rather than proceeding. Second, an anchor that implies θ above 1 means the published count exceeds the survey noncitizen count for that year and place, which the identity cannot produce; the command warns, and the usual cause is a mismatched geography or vintage between the anchor and the ACS series.

[4] Install with `net install residualundoc, from(https://raw.githubusercontent.com/ericabooth/residualundoc-replace force`. The package is a convenience and is not required to reproduce any result here, since the replication scripts stand alone.

9 Implications for Applied and Policy Work

We built the method for one kind of user: an analyst who needs a current, defensible number for a state and has neither the time nor the inputs to rebuild a demographic model before a hearing or a budget cycle. For that user it does a few useful things. It produces a figure the moment the ACS posts and updates itself when the next benchmark appears, rather than waiting on a source's release schedule. It brings the choice of source into the open: any single undocumented-population number depends heavily on who produced it, and reporting the quick estimate next to a published state residual turns that usually invisible dependence into a stated range. And it carries over to program-relevant subtotals, as the insurance figures show, using nothing more than a published subgroup table and one benchmark rate. Most useful of all, the method flags where it is wrong: a national ratio is neither good nor bad in general; it is close through most of the country and predictably off in a handful of identifiable states, and the map is what lets an analyst tell the two apart before quoting a number.

The policy payoff is narrower than the method, but concrete. State budget and health agencies routinely need the size and coverage of a population they cannot survey directly: undocumented residents are ineligible for most federally funded insurance, so they appear in uncompensated-care costs, emergency Medicaid caseloads, and community health center demand. The pieces this paper assembles map onto those planning tasks. A current-year state total sizes the affected population before a session; the uninsured split (1.3 to 1.4 million in Texas, 45 percent nationally) sizes the uncompensated-care exposure; the state uninsured map shows where that exposure concentrates; and the subgroup extension locates it, for example among families with young children. These are descriptive inputs carrying the caveats above, not causal or predictive claims, and which policies a state adopts in response is outside this paper's scope. For the recurring, mundane task of putting a defensible number on a hard-to-count population before a deadline, a transparent shortcut that shows its own error is often more useful than a more elaborate estimate that arrives too late or cannot be checked.

10 Limitations

A few cautions bound all of this. The most basic is that legal status is modeled, not seen: the ACS noncitizen count folds together lawful permanent residents, temporary legal migrants, refugees, asylees, people with liminal status, and undocumented residents, and θ is the only thing separating the last group from the rest. Because the method takes θ from a published anchor instead of rebuilding it, it inherits that anchor's accuracy and does not re-derive the admissions, mortality, emigration, or undercount pieces behind it. Section 5.4 sizes that inherited step, and the answer is sobering: the coverage reconciliation accounts for roughly half the published total, every estimate here rests on it, and we do not test it. The 2024 comparison should therefore be read as a benchmark against one full residual construction rather than validation against an observed count, since both quantities start from the same ACS noncitizen count. Because the anchor sources scale their survey counts up for undercount before dividing, θ is not literally the undocumented share of the raw ACS count, and any share taken against unadjusted ACS population is approximate. The all-states shortcut is, by construction, a first pass, and the smallest states are the noisiest, both because CMS rounds their counts to the nearest thousand and because the ACS samples them thinly. Finally, the uncertainty we report is deliberately partial: we carry ACS sampling error as a confidence interval and cross-source disagreement as a separate range (Section 5.5), but we do not propagate the undercount adjustment or the insurance-rate calibration into those bounds, and a single fully probabilistic interval combining every modeled step remains the obvious next step (Van Hook et al., 2021; Goes and Engelhardt, 2026).

11 The Same Identity Reads as a Time Series: A Texas Sub-Analysis

The same identity supports a time series. Instead of a national θ across states in one year, we fix θ at each year a source publishes and interpolate between anchors, so the estimate reproduces the published history and lets the annual ACS count carry the years in between. Figure 9 shows this for the United States and Texas, 2008 to 2024, anchored on Pew nationally and MPI for Texas, with the published values marked. The estimate declines from 11.4 million nationally in 2010 to 10.2 million in 2019, then rises to 14.9 million by 2024, and the Texas line reaches 2.1 million. Running the same construction on CMS anchors gives a second, independent vintage; the two converge on a 2024 national total near 14.6 to 14.9 million. Two elements lie outside the scope of the sub-analysis and are worth flagging for anyone extending it: the methodology break that Pew and CMS both introduced in their 2022–2023 revisions, and the odds-scale insurance transport for years without a published split.

12 Conclusion

The residual method does not have to be heavy to be useful. Its central quantity, the undocumented share of noncitizens, moves slowly enough that a published anchor and an annual survey table reconstruct state counts and reproduce the estimates they rest on. Applied to all states for 2024, the shortcut matches the CMS residual within 10 percent in 22 of 51 states and within 20 percent in

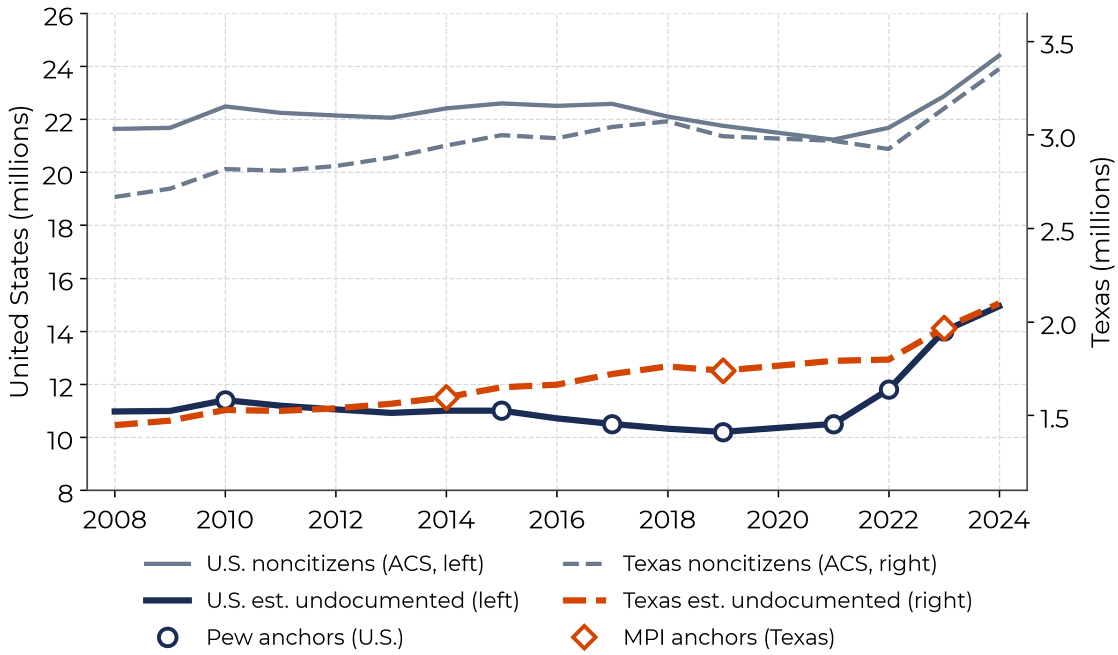

Figure 9. Time-trend version of the method, United States and Texas, 2008–2024. Hollow markers are published Pew (U.S.) and MPI (Texas) anchors; lines pass through them and interpolate ACS movement between. The two vertical axes differ: United States on the left, Texas on the right. The Texas count (about 2.1 million in 2024) is roughly a seventh of the U.S. count (about 14.9 million); the lines sit close on the page only because they are drawn on different axes and are not directly comparable in level.

most states. It misses in a predictable, mapped set of places, missing high where legal immigrants dominate the noncitizen population and low in the newer Southern destinations. For an analyst who needs a current, transparent, reproducible state number on a policy timeline, that is often enough, and the difference map says exactly when it is not.

Data Availability

All inputs are public and were retrieved in July 2026. The replication package (Stata do-files, a Python mirror, input and output CSVs, an anchor log, and a readme) reproduces every figure and table.

- U.S. Census Bureau, ACS 1-year detailed tables B05001 (nativity and citizenship) and B27020 (health insurance by citizenship), 2024, via the Census Data API: `https://api.census.gov/data/2024/acs/acs1` (table shells `https://api.census.gov/data/2024/acs/acs1/groups/B05001.html` and `.../B27020.html`). Accessed 2026-07-03.
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